Assignment 3: Ant Colony

Multi-Agent System

LINFO1361 – Artificial Intelligence – 10th Mar

Assignment 3

General objectives

Implement three different agents to simulate the behavior of ants

Assignment 3

General objectives

Implement **three different agents** to simulate the behavior of ants

1. **Non-Cooperative Agent**
2. **Cooperative Agent**
3. **Smart Agent**

What is an Agent in an Ant Colony

An agent is an ant.

What is an Agent in an Ant Colony

An agent is an ant.

This means that it has to incorporate **random movement while looking for food.**

What is an Agent in an Ant Colony

An agent is an ant.

This means that it has to incorporate **random movement while looking for food.**

But it **cannot be completely random.**

What is an Agent in an Ant Colony

An agent is an ant.

This means that it has to incorporate **random movement while looking for food.**

But it **cannot be completely random**.

Depending on the agent, you will implement different strategies.

Non-Cooperative Agent

- For this agent, you have to implement an Ant agent that **cannot communicate** in any way with other Ants.

Non-Cooperative Agent

- For this agent, you have to implement an Ant agent that **cannot communicate** in any way with other Ants.
- However, this Ant agent **have a memory** and can register of all elements it perceived.

Non-Cooperative Agent

- For this agent, you have to implement an Ant agent that **cannot communicate** in any way with other Ants.
- However, this Ant agent **have a memory** and can register of all elements it perceived.
- ***Hint:*** *You could for instance map all the visited tiles and use algorithms you saw in the lectures*

Cooperative Agent

- For this agent, your agent **does not have any memory** but can **communicate** using **pheromones**.

Cooperative Agent

- For this agent, your agent **does not have any memory** but can **communicate** using **pheromones**.
- There are **two types** of pheromones:
  - Food pheromones
  - Home pheromones

Cooperative Agent

- For this agent, your agent **does not have any memory** but can **communicate** using **pheromones**.
- There are **two types** of pheromones:
  - Food pheromones
  - Home pheromones
- Those pheromones can be emitted using the framework we give you, but it’s up to you to decide when to drop them.
- ***Note:*** *For those who don’t know, a pheromone is a chemical signal that can be drop by animals to signal something to other animals.*

Smart Agent

- For this agent, it’s up to you to implement it the way you’d like.

Smart Agent

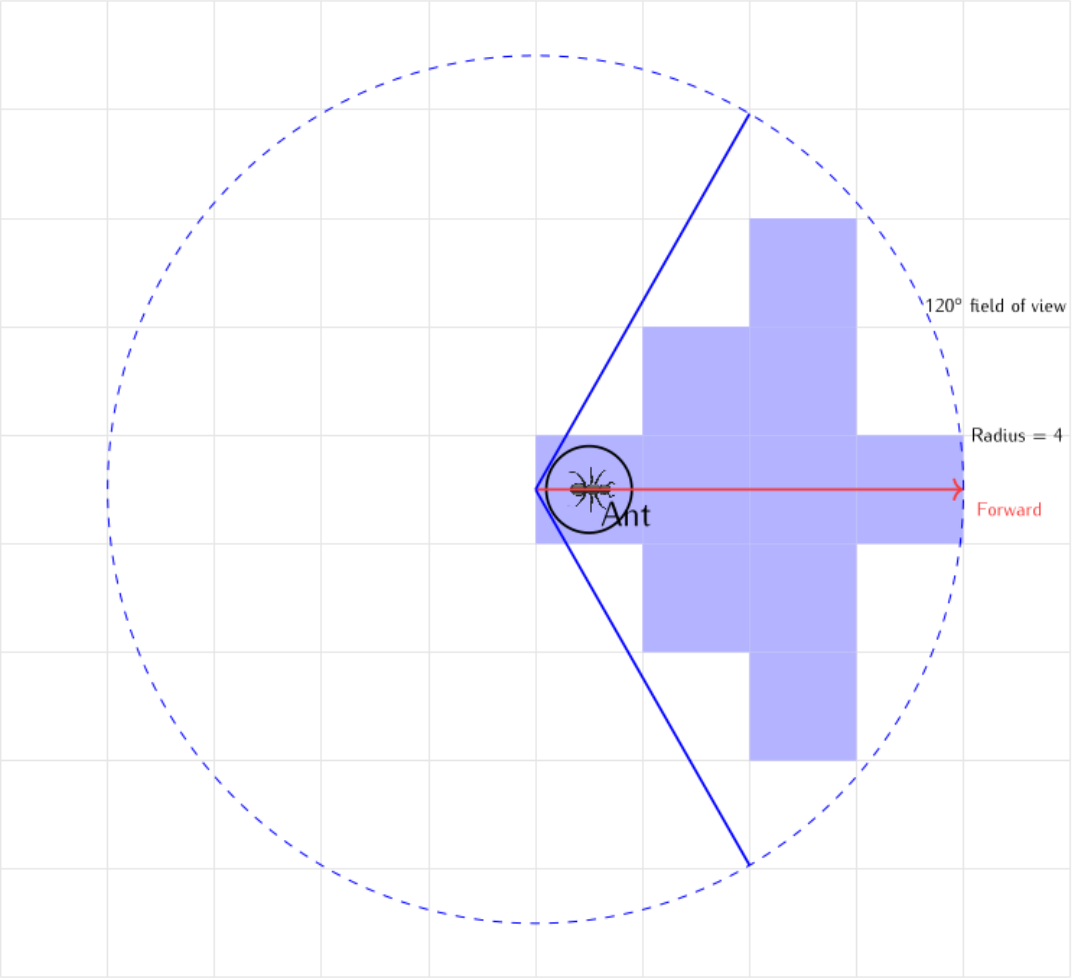
- For this agent, it’s up to you to implement it the way you’d like.
- It can **both** use **pheromones** and have a **memory**.

Smart Agent

- For this agent, it’s up to you to implement it the way you’d like.
- It can **both** use **pheromones** and have a **memory**.
- You can also try to implement **methods** you have **seen in the lectures** to make it more efficient.

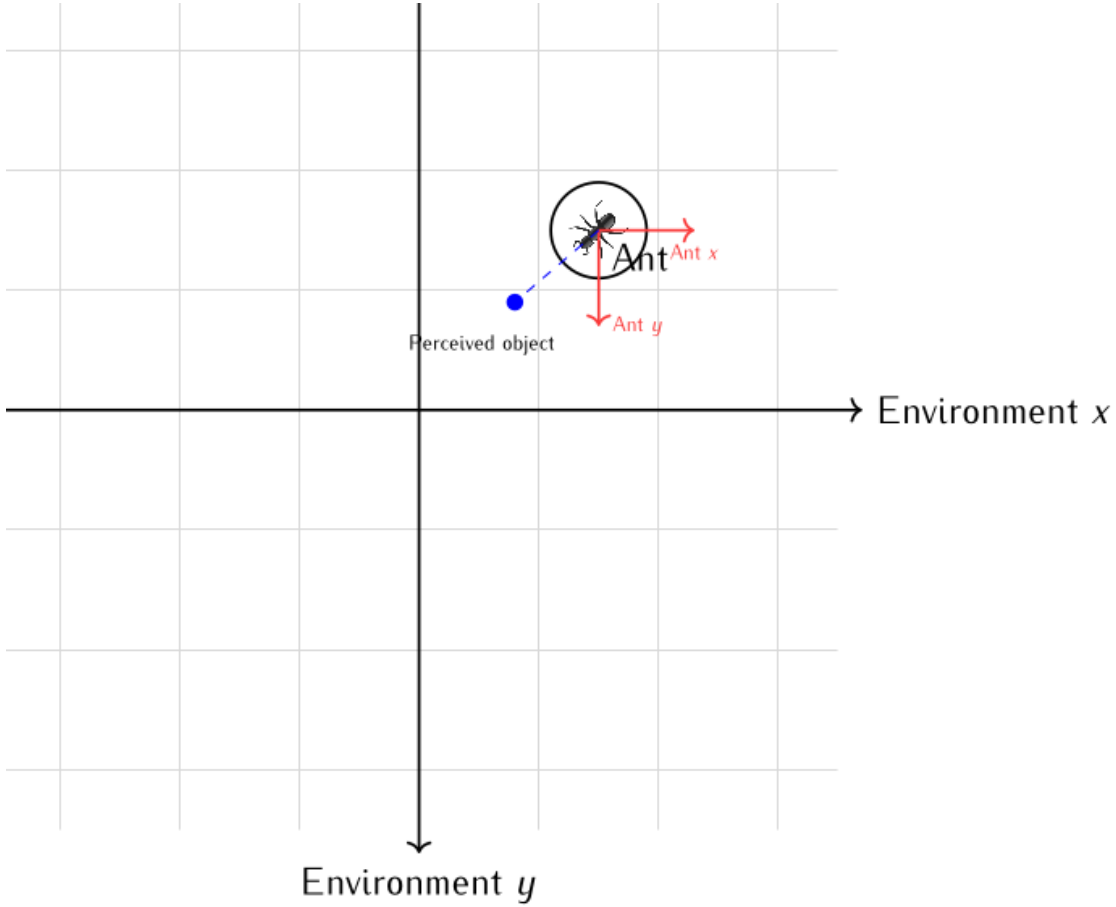
Framework Description

Ant Module



Framework Description

Ant Module



Framework Description

Evaluations

Both **non-cooperative** and **cooperative agent** will be evaluated on the basic environment called **XX_square_diag.txt**

Your **smart agent**, however, will be evaluated on all instances we have generated for this purpose (20). We provide you with 6 of them to test your model. The others have similar degrees of difficulty.

Finally, you will have to answer to **3 open questions**. We highly recommend you to use concepts seen in your lectures for this purpose.

Framework Description

Evaluations

Both **non-cooperative** and **cooperative agent** will be evaluated on the basic environment called **XX_square_diag.txt**

Your **smart agent**, however, will be evaluated on all instances we have generated for this purpose (20). We provide you with 6 of them to test your model. The others have similar degrees of difficulty.

Finally, you will have to answer to **3 open questions**. We highly recommend you to use concepts seen in your lectures for this purpose.

NOTE : You can also build your own environments. Details on how to do so are provided in the assignment document.

Open question

Evaluations

Q1 — Nombre de fourmis

- Objectif : atteindre 75% de nourriture collectée et 33% rapportée à la colonie.
- Variation : 1 → 200 fourmis, 10 runs par point.

Q2 — Taux d’évaporation des phéromones

- Paramètre : 0.5 → 0.999, 70 fourmis, 10 runs par point.

Q3 — Cheater strategy et analyses avancées

- Implémentation : accès complet à l’environnement, algorithme de chemin optimal

Framework Description

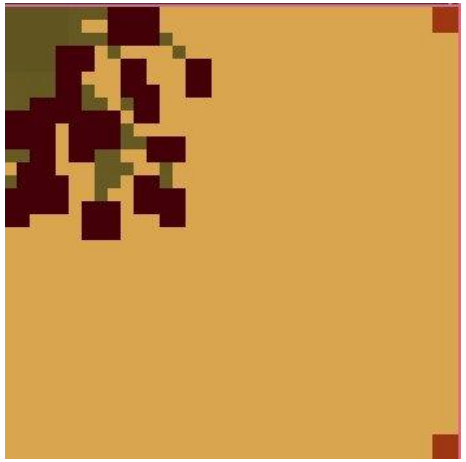
Evaluations

01_line_env

Framework Description

Evaluations

01_line_env

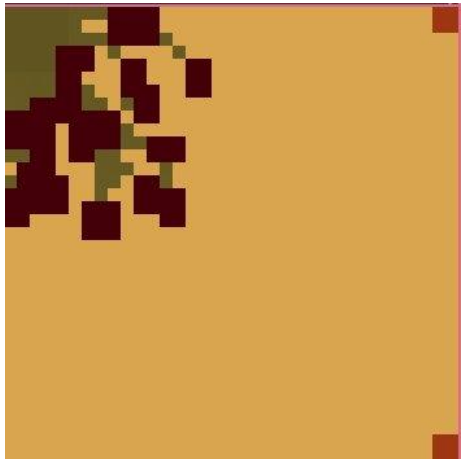


03_square_two_foods

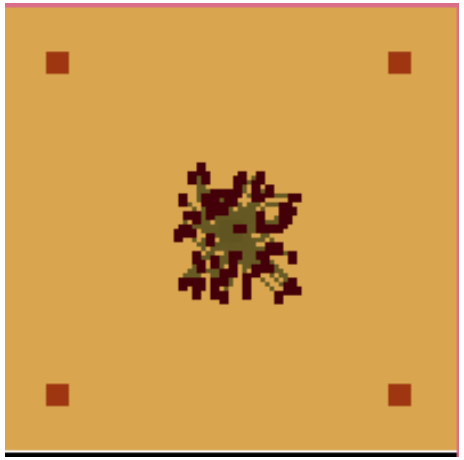
Framework Description

Evaluations

01_line_env



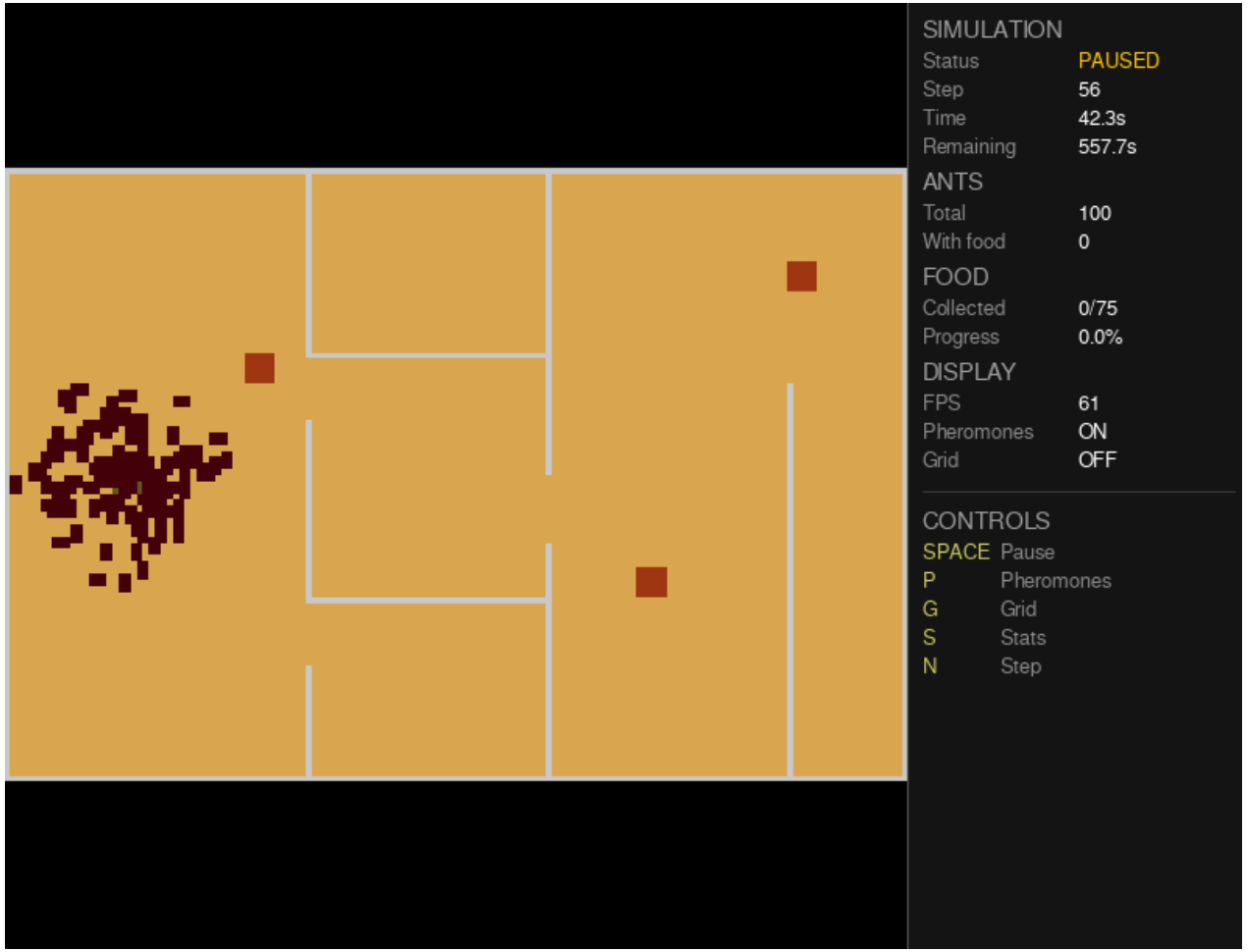
03_square_two_foods



05_square_four_foods

Framework Description

Evaluations



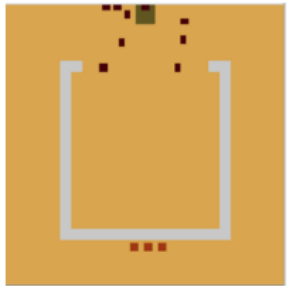
06_river_crossing.txt

Inginious

Auto-evaluation

The maximum time for all completions is 250 seconds.

See the environment:



Your answer passed the tests! Your score is 100.0%. [Submission #69c3e5430cf49ea44b60696f (**2026-03-25 14:38:11**)]

Your strategy	Status	Completion	Time (s)	Steps
inginious_ants1	Success	100.0%	3.6 / 50	34402 / 100000
inginious_ants5	Success	100.0%	3.5 / 50	8889 / 100000
inginious_ants10	Success	100.0%	5.0 / 50	6813 / 100000
inginious_ants15	Success	100.0%	4.5 / 50	3897 / 100000
inginious_ants50	Success	100.0%	6.9 / 50	1865 / 100000

Inginious
Strategy

SmartStrategy	Status	Completion
01_line_env	✅ Success	100.0%
02_square_env	✅ Success	100.0%
03_square_two_food_spots	❌ **Failed**	50.0%
04_obstacle_env	❌ **Failed**	75.0%
05_square_four_food_spots	✅ Success	100.0%

Framework Description

Evaluations

The grading will be the following:

- **Implementation** [14/20]
  - Non-cooperative agent [2/20]
  - Cooperative agent [2/20]
  - Smart agent: 0.5 point/environment [10/20]
- **Report** [6/20]
  - Open question 1 [2/20]
  - Open question 2 [2/20]
  - Open question 3 [2/20]